Manufacturing Dispersion: How Data Cleaning Choices Affect Measured Misallocation and Productivity Growth in the Annual Survey of Manufactures

by

Hang Kim
University of Cincinnati

Martin Rotemberg
New York University

T. Kirk White
U.S. Census Bureau

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Abstract

Measurement of dispersion of productivity levels and productivity growth rates across businesses is a key input for answering a variety of important economic questions, such as understanding the allocation of economic inputs across businesses and over time. While item nonresponse is a readily quantifiable issue, we show there is also misreporting by respondents in the Annual Survey of Manufactures (ASM). Aware of these measurement issues, the Census Bureau edits and imputes survey responses before tabulation and dissemination. However, edit and imputation methods that are suitable for publishing aggregate totals may not be suitable for estimating other measures from the microdata. We show that the methods used dramatically affect estimates of productivity dispersion, allocative efficiency, and aggregate productivity growth. Using a Bayesian approach for editing and imputation, we model the joint distributions of all variables needed to estimate these measures, and we quantify the degree of uncertainty in the estimates due to imputations for faulty or missing data.

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I Introduction

Dispersion is a pervasive characteristic in plant-level data. Some plants are big, others are small. Some look more productive than others, with relatively high shipments given their inputs. Even within industries, plants use different mixes of inputs. There is also dispersion across time, as some plants grow faster than others. Across a variety of fields, models are used to describe the causes and consequences of these differences. For instance, over the past twenty years, many economists (including us) have written papers and lecture notes highlighting that the within-industry misallocation of factors can help explain cross-country differences in productivity (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009). This paper describes some challenges with the measurement of dispersion in both levels and the growth rates of the productivity of U.S. manufacturing plants.

The confidence we have in our claims about dispersion in establishment[1] behavior depends on how worried we are about measurement error (Bartelsman et al., 2021). While measurement error has recently been recognized as an important concern for estimating cross-sectional misallocation (Bils et al., 2021; Velayudhan, 2018; Gollin and Udry, 2019; Esfahani, 2019; Rotemberg and White, 2021a), it has long been understood that measured changes within firms over time may be particurally sensitive to measurement error (Griliches and Mairesse, 1984). In principle, measurement error has an ambiguous effect on measured dispersion, since nonclassical noise can either push establishments' reported values towards or away from what is typical in their sector. In this paper, we focus on two potential sources of bias in estimates of dispersion in productivity levels and productivity growth: establishments potentially misreporting their own characteristics and subsequent editing and imputation procedures.

[1] Throughout the paper, we use the words "plant" and "establishment" interchangeably. An establishment is not necessarily the same as a firm. Some firms own multiple establishments.

Most statistics agencies initially ask establishments to verify (or send in) information, but the subsequent steps vary across surveys. Many statistical agencies in developed countries, including the U.S. Census Bureau, both edit and impute responses. The exact procedures vary across industries and time (White, 2014), but broadly take two forms. First, the Census Bureau *edits* some outliers. If a reported variable fails one or more edit rules, then it may be temporarily replaced with a missing value. Second, the Census Bureau *imputes* missing information, using other information reported by the plant (both in that year and, when available, in previous years) and other plants in the same industry.[2] For 2009-2018 we have access to the original values and cleaned values for plants in the Annual Survey of Manufactures, as well as the relevant edit flags.

This paper aims to evaluate the importance of misreporting, editing, and imputation in the ASM for measuring within-industry dispersion in the levels and growth rates of plant-level variables. We structure our analysis in three steps. First, we provide evidence of measurement error in the 2009-2018 ASM plant-level data, both in terms of levels and growth rates.

Second, we demonstrate the extent of data cleaning efforts undertaken by the U.S. Census Bureau and the extent to which this cleaning affects standard measures of productivity dispersion and allocative efficiency. Although the existing literature has examined the effects of imputation in the Census of Manufactures (Collard-Wexler and De Loecker (2020); White et al. (2018)) we are the first to make use of the raw reported data to show the full extent to which Census data cleaning affects measures of within-industry dispersion in both levels and growth rates.[3]

Finally, as an alternative to Census Bureau editing and imputation, we apply a re-

[2] Editing a variable normally involves deleting the original response, and then treating it as if it had never been reported.

[3] White et al. (2018) in particular focused on the effect of the Census Bureau's regression imputes on measured within-industry productivity dispersion. We go beyond just looking at regression imputes and show how all types of Census Bureau editing and imputation affect our statistics of interest.

cently developed Bayesian method for simultaneous editing and imputation, which has been shown to do a good job of preserving joint distributions in the data while satisfying standard editing rules like those the Census Bureau uses (Kim et al., 2015). We do not claim that this method is superior to all other data cleaning methods in all contexts.[4] Neither do we claim that our imputations are the best possible imputations for the U.S. Census data.

While the Bayesian editing and imputation model we use in this paper does not take advantage of the entire set of resources available to the U.S. Census Bureau, it nevertheless has some advantages over some of the Census Bureau's imputation methods. First, unlike the Census Bureau's imputations, which are fixed once tabulations of the data have been published, we take advantage of the full 5-year panel of ASM data—we use data from years $t+1$ to $t+4$ to impute for missing data in year t and vice versa. About 28% of plants in the ASM data have at least one of materials or total value of shipments imputed using the predicted value from a regression, a method known to understate the variance in the data (Van Buuren, 2018). Both for the edits and imputes, the resulting dispersion in the final ASM data is smaller in the regression-imputed data than for the rest. This is less true for the data imputed following Kim et al. (2015). Because it is a multiple imputation method, our Bayesian method can also quantify the uncertainty in our estimates due to missing data, unlike the Census Bureau's imputations in the ASM. Finally, the edits/imputations which have some of the largest impacts on our statistics of interest from the ASM are analyst edits, which are somewhat opaque and not replicable. Our method is both transparent (in the sense that we can provide the code to other researchers) and replicable across contexts.

[4] In general, the suitability of any imputation model depends on how the data will be used (Van Buuren, 2018). For example, univariate regression imputes such as those the Census Bureau frequently uses may be fine for tabulating industry totals, but they do a poor job of preserving within-industry total factor productivity dispersion (White et al., 2018).

Economists have a long tradition of studying (mis)measurement (Frisch, 1934; Griliches, 1974). Our paper is firmly in the spirit of Romer (1986a,b, 1988, 1989) and Balke and Gordon (1989), who study how differences in data quality matter for understanding the historical incidence of business cycles and unemployment (Burns, 1960).[5] We also complement current efforts to understand measurement error in labor force surveys (Abowd and Stinson, 2013; Kambourov and Manovskii, 2013; Meyer and Mittag, 2019; Medalia et al., 2019; Vom Lehn et al., 2020). In the misallocation literature, Gollin and Udry (2019), and Esfahani (2019) study the role of measurement. Their approaches use economic theory to distinguish between measurement and "true" misallocation, for instance by arguing that farmers are unlikely to misallocate resources across their own plots, or that firms misreport to avoid additional taxes. Bils et al. (2021) explicitly studies measurement error in manufacturing, leveraging theory and panel data in order to argue that measurement error is increasing in the U.S. (using the already-processed final data).[6] One value of our approach is we provide direct evidence of measurement error. Furthermore, researchers can use our proposed data cleaning procedure regardless of their question, and even in settings with less extensive data collection and cleaning efforts than in the ASM.

II Data Description

We primarily use micro-data from two five-year panels of the U.S. Annual Surveys of Manufactures (ASM) (U.S. Census Bureau, 2018), augmented with capital constructed by Blackwood et al. (2023): 2009-2013 and 2014-2018. We describe the data sources in more detail in Appendix Section A. Like nearly all previous research using the ASM, we use the data the Census Bureau provides to researchers in the Federal Statistical Research

[5] Similar issues have recently been discussed in the asset pricing literature, since measuring the correlation of consumption with asset prices is difficult (Savov, 2011; Kroencke, 2017). Data from international tests also can be imputed (Jerrim et al., 2017; Troccoli, 2020).

[6] The intuition of their theoretical result is that measurement error lowers the correlation of *changes* in inputs and revenue, which they can then use to correct the naive cross-sectional measurement of misallocation. The captured data for the annual panel surveys that Bils et al. (2021) use are not available, so we cannot speak directly to their results.

Data Centers after the Bureau has done extensive cleaning of the data. Unlike any other research that we are aware of (other than previous versions of this paper), we also use the raw data as it was reported to the Census Bureau by the surveyed plants.[7]

The captured data differs from the final data in two respects. First, missing values due to non-response in the reported data are imputed in the cleaned data, using a variety of imputation strategies. Second, responses which fail edit rules in the reported data are normally imputed or changed in the final data.

We focus on four variables used to measure plant-level total factor productivity: total value of shipments, total cost of materials (which includes the cost of energy), total wages, and the capital stock. The first three variables are directly measured in the ASM, and are present in both the raw data and the final data. Capital is more difficult to measure: the plants report fixed assets at the beginning and end of the year, and only in years ending in 2 or 7 (Economic Census years); but for productivity what matters is the flow of capital services. In the final data we use the measures of real stock of capital carefully constructed by Blackwood et al. (2023), multiplied by nominal rental rates to measure the cost of capital services (Kehrig, 2015). In the raw data we use backwards and forwards perpetual inventory to construct our own measures of capital stocks from reported assets and capital expenditures.[8] We use the same nominal rental rates that we used for the final data to convert the raw capital stocks to costs of capital services.

III Evidence of Measurement Error in the ASM

Before describing the Census Bureau's data editing and imputation procedures, we provide some *prima facie* evidence of the *need* to clean the data. In the ASM, there are questions that ask about inventories at the end of the reference year vs. the end of the prior

[7] Throughout the paper we use "final data" and "cleaned data" or "Census-cleaned data" interchangeably. Likewise we use "raw data", "captured data", and "reported data" interchangeably.

[8] In robustness checks we find that these different measures of capital do not have much effect on measured productivity.

year.[9] The three types of inventories measured in the ASM are finished goods, materials, and work-in-progress. For plants that are surveyed in two consecutive years, we can check if the respondent reported the same value for end-of-year-t values in the year t and year $t+1$ surveys, and if not, how big is the difference. Figure 1 shows the share of observations in the ASM for which the reported values for the same plant and same reference year differ by more than 50%.[10] The shares are less than 5% for all three variables in four of the eight years. However, for 2010 more than 20% of the plants reported values that differ by more than 50% for all three variables. For 2015, more than 40% of plants did so.

For a few variables in the ASM, we can compare the reported values to IRS payroll or income tax records in the Census Bureau's Business Register. In particular, for annual payroll, the ASM instructions say:

> Report the annual payroll at this establishment, before deductions. This value should be the same value included on Internal Revenue Service Form 941, Employer's Quarterly Federal Tax Return.

Firms with multiple establishments may combine their payroll when reporting to IRS. However, for firms with only one establishment ("single-units"), the annual payroll reported to the ASM should be the same as the sum of the four quarters of payroll reported on IRS Form 941 for the same plant in the same year.[11] The ASM measure of employ-

[9] For example, item 9A on the 2018 ASM questionnaire asks: What was the value of inventories owned by this establishment as of December 31 before Last-In, First-Out (LIFO) adjustment (if any) for Finished goods (final output of this establishment, but still within ownership) at the end of 2018 and at the end of 2017?

[10] We do not calculate the differences in the years 2009 and 2014 because they are start of 5-year ASM panels of smaller plants. For those years, half of the sample has no prior-year ASM data.

[11] In principle, differences between annual payroll reported to the ASM and annual payroll reported to the IRS for the same plant could be evidence of misreporting of payroll to the IRS and not necessarily to the ASM. Several facts go against this interpretation of the differences. There are substantial financial penalties for failure to report or inaccurate reporting of payroll tax liabilities to the IRS (see https://www.irs.gov/instructions/i941). There are no such penalties for misreporting payroll or any other variable on the ASM. Furthermore, the Census Bureau uses IRS annual payroll data as part of its editing and imputation process, sometimes substituting IRS annual payroll data for missing or implausible payroll data reported to the ASM.

ment is also similar to the employment measure from IRS Form 941. The ASM measure of the value of plant output—total value of shipments—is not necessarily the same as the revenue concept measured on IRS income tax forms, but for many single-unit establishments in the ASM, the values are similar. Table 1 summarizes the differences between payroll, employment, and shipments/revenue in the 2009-2018 ASM vs the IRS tax data. Panel A compares the variables in levels, and Panel B compares growth rates. On average 21% of the plant-year observations in our sample have ASM captured payroll that differs in absolute value by more than 10% from IRS payroll for the same plant-year, and 5% report values that differ by more than 50%. The analogous shares are similar for employment (28% and 5% respectively). The ASM reported value of shipments differs from the revenue measure from IRS income tax records by more than 10% for 25% of plant-years, and it differs by more than 50% for 15% of plant-years.[12] One might hope that large (in percentage terms) measurement error in the ASM is limited to smaller establishments. Unfortunately, at least for these single-plant firms, the shares of plants that have measurement gaps of 10 or 50% are similar across the employment size distribution. Panel B of Table 1 shows the shares of plant-years for which the annual growth rates in the ASM vs. the IRS data differ in absolute value by more than 10 percentage *points* (column 2) or more than 50 percentage points (column 3). The samples for Panel B are restricted to single-unit establishments for which there are reported values for the given variable in two consecutive years in both the ASM and the IRS data. Although the samples are different, the shares are similar to those in Panel A.

[12] Note that here we are comparing data from IRS tax records to the original data reported to the ASM. For annual payroll, and, to a lesser extent, employment, the Census Bureau uses the IRS tax records to impute for ASM data that it deems erroneous. Indeed, for single-plant firms with 50 or more employees, fewer than 1% have *final* ASM payroll values that differ from the IRS payroll value by more than 50%. However, even among the largest single-plant firms (those with more than 250 employees), the final ASM value of employment differs from its IRS tax records counterpart by more than 10% for 27% of the plant-year observations.

IV Data Cleaning in the ASM

As in most surveys, not all respondents to the ASM answer all of the questions, and—as Figure 1 and Table 1 show—some responses are inconsistent with each other or inconsistent with administrative records data from the same firms. The Census Bureau has created imputation and edit rules for this data, the development of which are described in Sigman (1997) and Thompson and Sigman (1999). There are essentially three types of edit rules: range edit rules, ratio edit rules, and balance edit rules. Range edit rules restrict the range of single variables. For example, employment cannot be negative. Ratio edit rules are upper and lower bounds for ratios of plant-level variables. These are industry-specific and can change over time. For example, the ratio of annual payroll over employment is not allowed to be greater than some number UB_j or less than some number LB_j for a plant in industry j. Balance edit rules require certain sets of variables to add up to other variables. For example, the sum of production worker wages and non-production worker wages should equal total wages. A response for a given variable or set of variables is said to "fail" an edit rule if it does not satisfy the inequality constraint or (in the case of balance edits) the equality constraint that defines the edit rule. The Census Bureau replaces edit-rule-failing responses and also imputes for data that is missing due to nonresponse.

Until a few years ago, it was difficult for researchers to identify which, if any, responses for a given plant were imputed.[13] In this paper we use the Census Bureau-provided data flags to identify edited and imputed data. We also go beyond the imputation flags and use the original responses from the establishments themselves.

[13] There is a tradition of researchers trying to back out imputations from the final data, see for instance Davis and Haltiwanger (1991), Roberts and Supina (1996), Syverson (2004a,b), and Collard-Wexler (2011). It was easiest to identify "hot deck" imputes, which led to duplicated observations. There are no hot deck imputes in the data we use, but item-level edit/impute flags for the 1987 and later censuses are available to researchers in the Federal Statistical Research Data Centers (FSRDCs) (White, 2014). It is worth noting that when, for example, Hsieh and Klenow (2009) was published, neither imputation flags nor the original plant responses were available.

The Census Bureau imputes for missing or edit-rule-failing data using a variety of methods, which are listed in Table A.1. There are too many types of edit categories for us to describe the effects of them individually (plus several are fairly rare), and a given plant-year observation may be subject to multiple edits or imputations for different variables. To give an idea of the overall exposure of plants to edits and imputations in the ASM, we group them into seven categories in Table 2. Half of plant-year observations have imputations for missing values in at least one variable needed to compute revenue total factor productivity (TFPR).[14] About 46% of plant-years have a positive final value for shipments, payroll or materials that is different from its (non-missing) captured value; for capital, the analogous share is more than half.

V The Effects of Census Bureau Data Cleaning on Measures of Dispersion and Growth

Section IV showed that data cleaning by the Census Bureau affects a large percentage of the observations in the ASM. Even so, the Census Bureau data cleaning in the ASM might not be a problem if it does not have much effect on statistics of interest. Alternatively, the cleaning might have a large effect, but still allow for valid statistical inference about the population parameters we care about. In this section, we show that the Census Bureau's data cleaning does have large effects on several frequently-studied measures of within-industry dispersion in levels and growth rates, as well as measures of plant-level persistence and aggregate productivity growth. We also explain why the Census Bureau's methods of imputation in the ASM do not allow for valid statistical inference about population counterparts of these measures.

First we look at the effects of data cleaning on some cross-sectional summary statistics, using different versions of the data. The first row of Table 3 shows estimates from the captured data, and the second row shows estimates from the final data. The samples

[14] White et al. (2018) found even higher rates of imputation for missing data in the 2002 and 2007 Censuses of Manufactures. Note that the ASM oversamples larger plants, and the Census Bureau puts more effort into getting responses from larger plants.

differ because of missing values in the captured data. The third row presents estimates from the final data, but using the same sample as row 1. (We discuss the final two rows of the Table in Section VI.B.) The first column shows the within-R^2 from a regression of the total value of shipments on annual payroll and industry fixed effects, and the second column shows the within-R^2 from a regression of shipments on payroll, the cost of materials, capital, and industry fixed effects. In both regressions, the right-hand-side variables account for much more of the variation in the value of shipments in the final data than they do in the captured data. Columns 1 and 2 of the third row show that the difference in the R^2s in the captured data vs. the final data is *not* driven by sample selection. The stronger correlations in the final data are caused by changes the Census Bureau makes to the captured data.[15]

The last three columns of Table 3 show how Census Bureau data cleaning affects measures of within-industry dispersion in ratios that are related to total factor productivity. Column 3, labeled τ_K, is the standard deviation of the ratio of plant-level shipments over the plant-level capital stock, multiplied by the industry-level share of capital's cost as a share of total costs, scaled by the industry-level weighted average of this ratio. In the Hsieh and Klenow (2009) model of the effects of misallocation on aggregate productivity, this is a summary of the degree of idiosyncratic distortions to the marginal revenue product of capital. Columns 4 and 5 show the analogous statistics for shipments over payroll (τ_L) and shipments over materials (τ_M). Across the columns, there is roughly 6 to 23 times more dispersion in the captured data than in the final data.[16] Comparing the dispersion measures in rows 2 and 3 suggests that the much lower dispersion in the final data is driven primarily by changes made to the captured data, not by sample selection

[15] Although the Census Bureau uses many imputation methods in the ASM, as we saw in Table 2, using the predicted value from a linear regression is one of the more common ones. This imputation method tends to bias correlation estimates upwards (Van Buuren, 2018).

[16] To facilitate comparison to the literature on misallocation, we have trimmed (dropped from the sample) the 1% tails of the distributions of τ_K, τ_L, and τ_M.

bias due to missing data.

Table 4 shows the effects of editing and imputation on within-industry productivity dispersion and two seminal models that connect productivity dispersion and aggregate productivity, using the same samples and versions of the data as in Table 3. The first column shows the standard deviation of the natural log of plant-level revenue total factor productivity (TFPR), scaled by industry-level TFPR, which is itself the revenue-weighted harmonic mean of plant-level TFPR.[17] There is more than twice as much within-industry dispersion in log TFPR in the captured data vs. the final data, and almost none of this is driven by sample selection. Column 2 shows average plant-level productivity as a share of aggregate productivity, as defined in Olley and Pakes (1996):

$$OP_{within-share} = \frac{\sum \frac{w_i}{\sum w_j} TFPR_i}{\sum \frac{w_i Y_i}{\sum w_j Y_j} TFPR_i}, \tag{1}$$

where w_i is the ASM sample weight for plant i, Y_i is total value of shipments from plant i, and $TFPR_i$ is plant i's revenue TFP.[18] The denominator is the Olley-Pakes definition of aggregate productivity, except that we use the ASM sample weights to make our estimates representative for the entire manufacturing sector. In captured data, average plant-level productivity accounts for six percentage points more of aggregate productivity than it does in the final data, and only one of these percentage points can be explained by sample selection.

[17] We scale by industry-level TFPR to make this measure comparable to the "relative" TFPR that feeds into sectoral productivity in the HK model—see Equation 3 in the Appendix. To estimate plant-level TFPR, we assume a constant returns to scale Cobb-Douglas gross output production function with capital, labor and intermediates as inputs. We use annual payroll as the labor input and the total cost of materials (including energy) for intermediates. We use the capital stock measures as described in Section II. We use industry-level cost shares to estimate the output elasticicties.

[18] Note that in the original Olley-Pakes (OP) formulation, the numerator is just the unweighted average productivity of all the plants, also called the "within" term. OP were using data from all plants in the Census of Manufactures in a particular industry, so they had data for their entire target population. Since we are using the ASM, we use the ASM sample weights to make our estimates representative for our target population, which is the entire manufacturing sector.

The final column of Table 4 shows the aggregate within-industry Allocative Efficiency measure developed in Hsieh and Klenow (2009) and Bils et al. (2021).[19] This measure can vary between 0 and 1. Assuming the Bils et al. (2021) model economy accurately describes the real economy, it measures aggregate productivity as a fraction of what aggregate productivity would be in a world with no within-industry distortions to marginal revenue products. According to this measure, the U.S. manufacturing sector is 15 times more productive in the final ASM data than it is in the captured data. Once again, very little of this difference is explained by sample selection.

While a large literature following Hsieh and Klenow (2009) has focused on cross-sectional dispersion statistics such as those in Tables 3 and 4, another literature has focused on both plant-level and aggregate measures of productivity *growth* (Baily and Hulten, 1992; Bartelsman et al., 2009, 2013; Olley and Pakes, 1996; Petrin and Levinsohn, 2012). Panel A of Table 5 shows how data cleaning affects autocorrelations of plant-level variables that are inputs to productivity measurement—shipments (Y), materials (M), capital (K), and payroll (L)—as well as revenue TFP itself. More precisely, each column shows the within-R^2 from a regression of each variable on its one-year lag and industry fixed effects. Across all five variables, conditional on industry, lagged values have more explanatory power (are more autocorrelated) in the final data than in the captured data. Panel B shows the effects of cleaning on the standard deviations of the annual growth rates of the first four variables. There is more dispersion in plant-level growth rates in the captured data than in the final data. This is especially true for capital, possibly reflecting the fact that capital has a higher rate of imputation. As we saw in the Tables of cross-sectional measures, very little of the difference between any of these dynamic measures—neither the regression R^2s nor the dispersion in growth rates—in the final data vs. the captured data is due to sample selection.

[19] See Appendix Section B for the formula and its derivation.

Table 6 presents the results of Petrin and Levinsohn (2012) [PL] aggregate productivity growth (APG) decompositions. APG is defined as the growth of aggregate value-added minus the growth of primary inputs weighted by their costs as a share of aggregate value-added.[20] All estimates in this Table are the time series averages of annual growth rates.[21] Column 1 shows the contribution of within-plant TFP growth ("technical efficiency growth") to APG. In the captured data, this term contributed 42 percentage points per year, but in the final data it was -1 percentage point. Column 2 shows the contribution of the reallocation of inputs to APG, and the final column shows PL APG. In all three columns the captured data show very large positive average annual growth rates, and the final data show small negative ones. Row 3 shows that little to none of the differences between rows 1 and 2 are explained by sample selection.

Across a variety of types of statistics related to productivity—static and dynamic, descriptive and modeled, plant-level and aggregate—Census Bureau data cleaning has an enormous effect. Given the evidence of measurement error presented in Figure 1 and Table 1, this might be both expected and necessary. However, not all imputation methods are created equal. About 28% of the observations in our ASM sample have at least one variable (shipments or materials) imputed using conditional mean imputation (predicted values from linear regressions) which is known to overstate the strength of the relationship between the imputed variable and the predictor variables (Schafer and Graham, 2002) and has been shown to significantly reduce measured within-industry productivity dispersion in the Census of Manufactures (White et al., 2018). Another issue is that the Census Bureau creates a single imputation for each missing or faulty datum. Single imputation does not allow the data user to quantify the uncertainty in estimates due to missing data.

[20] See Appendix Section C for the formulas.

[21] Since the non-certainty portion of the ASM sample is only a five-year panel—it leaves the sample after 5 years—we only calculate annual growth rates starting in the second year of each panel.

Finally, some types of Census Bureau data cleaning raise issues of transparency, replicability and comparability across studies, which recent research has emphasized is a problem in economics (Blanco-Perez and Brodeur, 2019; Chang and Li, 2015; Bartelsman et al., 2013). One important category of data cleaning in the ASM is analyst edits. These edits rely on the expertise of full-time industry specialists employed by the U.S. Census Bureau. Analysts can try to use alternative sources to verify or edit values, so these edits probably improve the quality of the data. On the other hand, these sources are not readily available to researchers using the ASM microdata, so these types of edits are neither transparent nor replicable. This lack of transparency and replicability almost makes cross-country comparisons more difficult to assess. Although edits by Census Bureau analysts edits are applied to only 9% of the observations in our sample, they account for 41% of the overall effect of Census Bureau data cleaning on (measured) Allocative Efficiency and 90% of its effect on aggregate productivity growth—see Table A.3 and Appendix Section D. In the next section, we describe an approach to editing and imputation which addresses these issues.

VI A Bayesian Approach to Cleaning Plant-level Data

In this section, we focus on the intuition behind a Bayesian editing and imputation algorithm and why we think this approach is particularly useful for large datasets with highly skewed, highly heterogeneous business data like the ASM. We also highlight aspects of the Bayesian method that contrast with the Census Bureau's current editing and imputation methods. We provide further details of the Kim et al. (2015) algorithm [for this section, KCKRW] and details of our implementation of it in Appendix E.

This method is transparent and replicable in the sense that we can provide the code to other researchers, allowing them to go from raw unedited data to final edited/imputed data. It avoids the well-known biases introduced by legacy Census Bureau methods such

as conditional mean imputation. Since it is a multiple imputation method, it allows the user to properly quantify the uncertainty in their estimates due to missing data.[22] Finally, it can be applied to raw microdata from different countries and different time periods, allowing for greater comparability.

For clarity, we begin with a bit of notation. We have data for a set of n establishments. Establishment i reports p characteristics, $\boldsymbol{y}_i = \{y_{i1}, y_{i2} \dots y_{ip}$ (where some items could be missing). The corresponding true values are $\boldsymbol{x}_i = \{x_{i1}, x_{i2} \dots x_{ip}$, with an underlying distribution f. s_{ij} indicates if response j for establishment i is incorrect. Given the dataset of reported values $\boldsymbol{Y} = \{\boldsymbol{y}_1, \boldsymbol{y}_2, \dots \boldsymbol{y}_n\}$ the goal of data cleaning in principle varies depending on the objective of the statistical agency or the researcher. For example, the goal of the statistical agency might be to produce unbiased estimates of industry totals of individual variables, cross-tabulated with geography and establishment size.[23] For studying productivity dispersion or productivity growth, it is important that our edited-imputed data reflect the underlying joint distribution of plant characteristics $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \dots \boldsymbol{x}_n\}$ and not just the marginal distributions. The Bayesian approach of KCKRW is designed to reflect the joint distribution while simultaneously insuring that the data satisfy internal consistency and plausibility checks provided by the statistical agency or the researcher. In particular, the KCKRW method ensures that ratios of a given plant's variables fall within industry-year specific bounds.[24]

The ratio bounds determine a feasible region for the data. If any of a plant's reported data falls outside the feasible region, then that plant's record is considered in

[22] Manski (2013) argues that properly quantifying uncertainty of estimates is an important feature missing from much public policy analysis of economic data.

[23] However, see Blackwood et al. (2023) for a description of an joint BLS-Census experimental data product which publishes industry level measures of productivity dispersion.

[24] The original KCKRW method also allows for constraints that certain sets of variables must add up—for example, the sum of production worker wages and non-production worker wages must equal total wages. We do not use any adding-up constraints in our application of the KCKRW method, because we only use the plant-level "totals" variables in our imputation models.

error. Once a record is determined to be in error, the next step is to determine which variable or variables in that record need to be corrected ("error localization"). In this step, the KCKRW algorithm takes advantage of information that the Census Bureau's error localization method does not. By simultaneously using the joint distribution of the edit-passing "good" data and the edit constraints a faulty record is failing, the KCKRW algorithm chooses probabilistically (a) which variables in the faulty record should be replaced with imputations and (b) what those imputations should be.

For example, suppose plant i reports $\{y_{i1}, y_{i2}, y_{i3}, y_{i4}\}$ where $\frac{y_{i1}}{y_{i2}}$ fail the ratios bounds. Depending on the joint distribution of $\boldsymbol{y}_i$, the KCKRW algorithm might replace both y_{i1} and y_{i2} with values that are highly likely given the reported values for y_{i3} and y_{i4}. The Fellegi and Holt (1976) method that the Census Bureau (and many other statistical agencies) uses seeks to minimize the number of edits, and ignores the joint distribution of the variables. In this example, if the Census could satisfy all the edit constraints by replacing only y_{i1} or only y_{i2} it would do so, without estimating the distributions of $\hat{f}\{x_{i1}|x_{i2}, x_{i3}, x_{i4}\}$ or$\hat{f}\{x_{i2}|x_{i1}, x_{i3}, x_{i4}\}$. One reason for wanting to minimize the number of changes to the data rather than preserve the joint distribution may be philosophical: the statistical agency may wish to keep as much of the originally reported data as possible.[25]

Once a decision has been made about which variables in a record are in error, the final step in the Census edit-imputation algorithm is to replace the variables in error with imputations.[26] In this step the Census Bureau uses a variety of methods, depending on what data is available for use in imputations. For example, for payroll, the Bureau has access to alternative data from IRS payroll tax records. For materials, no such administrative records data are available, so the Bureau most frequently uses the predicted value from

[25] Another reason may be that when the Fellegi and Holt (1976) method was developed, computational power and storage capacity was a small fraction of what it is today.

[26] Note that while these are separate steps in the Census algorithm, in the Kim et al. (2015) algorithm, error localization and imputation are done simultaneously.

a regression of materials on shipments to predict missing materials. The Census method is hierarchical, in the sense that it first tries the "best" method in its arsenal, and if that imputed value does not satisfy all the edit constraints, it moves on to the next method. One advantage of the Census method is that it is guaranteed to eventually produce imputations for every plant.

In contrast to the Census Bureau's approach, the KCKRW algorithm uses a statistical model of the joint distribution of the data for a given sample of plants (normally an industry or industry-year).[27] It does this using a Bayesian non-parametric approach. Rather than assuming the data come from any particular distribution, the approach uses a truncated Dirichlet process mixture of normals (Ishwaran and James, 2001). The algorithm takes a finite number of draws of the Dirichlet process, where each realization is itself a normal distribution with a particular mean and variance. Each of these realizations is a *component* distribution of the overall mixture distribution for the given sample. The number of components is determined probabilistically. This feature of the algorithm allows it to flexibly model many different types of distributions with little input from the modeller. For example, for a sample with a small number of observations that appear to come from a gaussian distribution, the algorithm might choose a single component distribution. For a sample with a large number of observations that appear to come from a highly skewed multi-modal distribution, the algorithm will draw multiple component distributions, each with different mean vectors and different variance-covariance matrices to fit the data.[28]

In our imputation models for the ASM data, we use 5-year panels with 5 year-specific

[27] One potential downside of the KCKRW approach relative to the current Census Bureau approach is that the KCKRW algorithm requires that at least two variables used in the imputation model are observed for every plant. (This does not have to be the same two variables for every plant.)

[28] There are of course still other researcher choices in the implementation described in Appendix E: for instance we build separate models for each industry-year, where a US industry is defined at the 4-digit NAICS level.

variables (employment, annual payroll, cost of materials, total value of shipments, and capital). Thus we model the joint distribution of the variables within an industry-year and over time. We then use the actual Census ratio bounds for those variables.[29] The Census Bureau also creates upper and lower bounds on the annual growth rates of ratios of variables in the ASM, and we use these bounds in our model as well.

VI.A Implementation of the Bayesian Approach

For each 4-digit NAICS industry-year in the ASM, we run a single chain of Markov Chain Monte Carlo with a burn-in of 2000 iterations. In the first step, we estimate a model of the joint distribution of the edit-rule-passing data. The second step takes draws from the model to fill in the missing data and edit-failing data. We then re-estimate the model parameters on the combined dataset for each iteration. Note that imputations will differ across iterations both because the draws for missing data are different and because the draws s_{ij} for which ratio-edit-rule-failing variables are in error may be different across iterations. For inference, after burn-in, for each industry-year we continue the chain another 5,000 iterations, and sample 10 implicates (i.e., completed datasets), keeping every 500th iteration.

VI.B Results using the Bayesian Editing and Imputation Approach

We estimate each of the parameters of interest targeted in Tables 3-6 separately on each of our 10 implicates. The fourth and fifth rows of each of the Tables shows the across-implicates mean and standard deviation, respectively, of these estimates. The standard deviation quantifies the uncertainty in our estimates due to missing data. We cannot quantify this uncertainty with the Census Bureau's single imputations.

Column 1 of Table 3 shows that in our Bayesian edited-imputed (EI) data payroll ex-

[29] The relevant ratios that are explicitly bounded (above and below) by Census are $\frac{shipments}{materials}$, $\frac{employment}{payroll}$, $\frac{shipments}{payroll}$, and $\frac{shipments}{assets}$.

plains a share of the within-industry variation in shipments that is very similar to the within-R^2 in the final data—0.84 vs. 0.85—and there is very little uncertainty about this estimate due to missing data. For the within-R^2 in column 2—from a regression of shipments on capital, labor, materials and industry fixed effects—we reach a similar conclusion. On the other hand, our Bayesian EI estimates of within-industry dispersion of the marginal revenue products of capital and materials are significantly higher than their corresponding estimates from the final data—nearly double in the case of capital.

Table 4 shows the effects of editing and imputation on our model-based cross-sectional statistics. The across-implicates mean of within-industry dispersion of (log) TFPR in our Bayesian EI data (0.51) is more than two standard deviations higher than the estimate from the final data (0.44), but still much lower than the estimate from the captured data (0.92). The Olley-Pakes share of aggregate productivity explained by (sample-weighted) average plant-level productivity is similar in the Bayesian EI data, and there is not much uncertainty about this estimate. However, the different methods of editing and imputation have a huge effect on the Bils et al. (2021) measure of Allocative Efficiency (AE): the mean estimate of AE in the Bayesian EI data is seven times higher than in the captured data, but less than half of the estimate from the final data. While there is some uncertainty in the Bayesian EI estimate of allocative efficiency, the estimate from the final data is roughly four standard deviations higher.

Panel A of Table 5 shows that in the Bayesian EI data, the within-R^2s from the panel regressions of shipments (Y), materials (M), and payroll (L) on their own lagged values are similar to the corresponding within-R^2s from the final data. However, the Bayesian EI estimate of the within-R^2s from the capital regression (0.81) and the TFPR regression (0.31) are significantly lower than their counterparts from the final data—0.97 for capital and 0.56 for TFPR. There is also considerable uncertainty about the Bayesian EI estimate for the TFPR regression: the across-implicates standard deviation is 0.09.

In Panel B we see that Bayesian EI estimates of dispersion in the plant-level growth rates of shipments, materials, capital, and payroll are always as high or higher than the estimates from both the final data and the captured data. For the case of capital, the Bayesian EI estimate of the standard deviation of plant-level growth rates (0.55) is more than double the estimate from the final data (0.16).

Finally, Table 6 shows the effects of our Bayesian EI method on the Petrin-Levinsohn aggregate productivity growth decomposition. In this case the Bayesian EI estimates are very similar to the final estimates for the Petrin-Levinsohn decomposition, although the standard deviation shows that there is considerable uncertainty around these estimates.

VII Discussion

There is a huge scope for different data editing and imputation choices to affect estimates of productivity dispersion and productivity growth in the U.S. manufacturing data. For example, different choices in editing and imputation change measured allocation efficiency in the raw data by a factor of 7 (our Bayesian cleaning) vs. a factor of 15 (Census cleaning). Lagged plant-level productivity explains 56% of the within-industry variation in productivity in Census Bureau-cleaned data, but only 31% of the variation in our Bayesian edited-imputed data. There is also considerable uncertainty in many of the estimates in this paper due to missing data.

Given the large internal inconsistencies we document in the data reported by some establishments and the significant rates of item non-response by other establishments, editing and imputation of the raw data is clearly necessary. However, it is important to note that imputation methods that are suitable for the primary goals of most statistical agencies—e.g., publishing industry or subindustry totals—may not be suitable for the goals of researchers using the microdata that the agencies collect. A long literature has documented the large degree of within-industry dispersion in productivity and pro-

ductivity growth in the manufacturing sector (Syverson, 2011). Researchers interested in explaining the causes and consequences of this dispersion need to understand the extent to which and the ways in which the manufacturing data itself is (and is not) "manufactured."

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VIII Figures and Tables

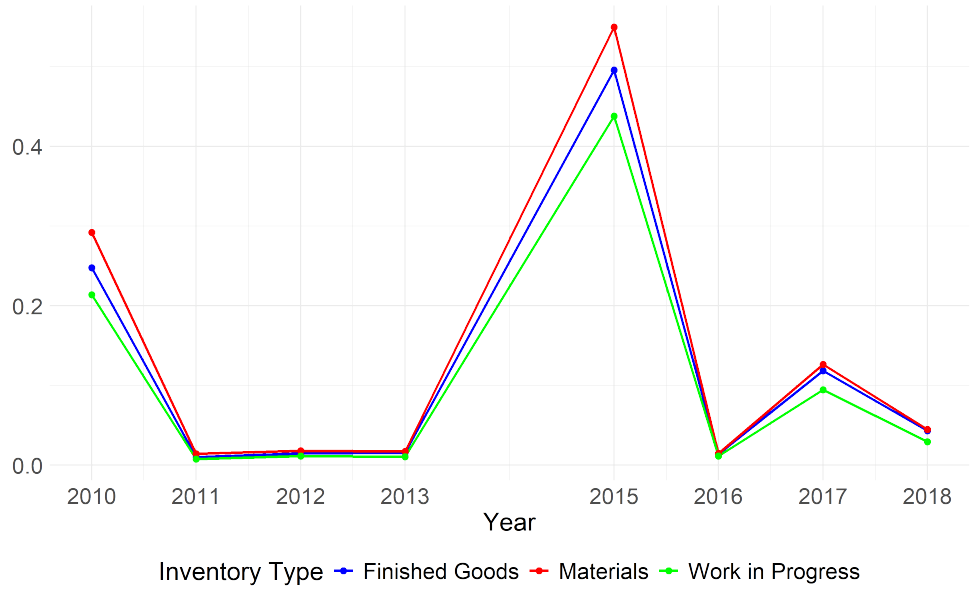


Figure 1: Inventory Measurement Error Over Time, ASM. Share of reported values for same plant-year that differ by greater than 50%. The figure shows the (unweighted) shares for finished goods inventories (blue), materials inventories (red), and work-in-progress inventories (green) from 2010-2013 and 2015-2018.

Table 1: Measurement Error in U.S. Annual Survey of Manufactures: Captured Data vs. Administrative Records

Panel A: Levels			
	sample size	Gap $>$ 10%	Gap $>$ 50%
Annual Payroll	121,200	0.21	0.05
Employment	101,200	0.28	0.05
Shipments/Revenue	116,200	0.25	0.15
Panel B: Growth Rates			
	sample size	Gap $>$ 10 pp	Gap $>$ 50 pp
Payroll Growth	77,000	0.26	0.06
Employment Growth	51,400	0.32	0.05
Shipments/Revenue Growth	80,800	0.24	0.11

Note: Panel A shows the share of observations in the 2009-2018 ASM for which the value reported to the ASM differed from the corresponding value reported to IRS by more than 10% or more than 50%. The samples are restricted to single-plant firms in the ASM that reported positive values for the given variable in both the ASM and the tax records. Panel B shows the the share of observations for which the annual *growth rate* of the given variable in the ASM differed from the corresponding growth rate in IRS data by more than 10 percentage points or more than 50 percentage points. The samples in panel B are restricted to single-plant firms in the ASM with reported positive values in two consecutive years in both the ASM and the IRS data.

Table 2: Extent of Editing/Imputation in U.S. Annual Survey of Manufactures

Edit/Impute Category	Share of Observations	Underlying Flags
Imputation for missing	0.50	Missing values (non-response) for shipments, materials, payroll, or capital were replaced by an imputation.
Any flow variable missing	0.30	Any of the variables shipments, payroll, or materials is missing in the captured data.
All flow variables missing	0.24	All three of the variables shipments, payroll, and materials were missing in the captured data and imputed with positive values in the final data.
Capital edit	0.53	Reported value for capital was replaced in the final data using a method other than analyst edit or the divide-by-1000 edit
Any edit except to capital	0.46	Reported value for shipments, payroll, or cost of materials was replaced with a positive value using any method.
Exactly 1 edit	0.54	The final value of exactly one of the variables (shipments, payroll, materials, or capital) was imputed (captured value could be missing or reported).
Any regression edit	0.28	Either shipments or materials is imputed using a linear regression.

Note: The table shows the extent of editing and/or imputation done by the Census Bureau in the Annual Survey of Manufactures. The 'Share of Observations' is the share of plant-year observations that fall into a given category, averaged over the 2009-2013 and 2014-2018 ASM panels. A plant-year observation can be in more than one category. Shares are unweighted.

Table 3: The Effects of Data Cleaning on Cross-Sectional Correlations and Within-industry Dispersion Estimates, U.S. Annual Survey of Manufactures, 2009-2018

Data Version	N	R^2: Y on L	R^2: Y on K, L, M	τ_K (s.d.)	τ_L (s.d.)	τ_M (s.d.)
		(1)	(2)	(3)	(4)	(5)
Captured	188,000	0.69	0.71	25.79	14.12	8.75
Final	364,000	0.85	0.86	1.44	0.63	1.54
Final*	188,000	0.84	0.86	1.52	0.62	1.59
Bayesian (mean)	366,000	0.84	0.85	2.72	0.63	1.77
Bayesian (s.d.)	366,000	0.00	0.00	0.23	0.02	0.12

Note: The table shows cross-sectional statistics from different versions of the 2009-2018 ASM data. The first row shows estimates calculated from the raw data, before any editing or imputation. The second row shows estimates from the final data after all Census Bureau editing and imputation. The third row shows estimates from the final data, but restricted to the same sample of plants as the first row. The fourth and final rows show the mean and standard deviation, respectively, of the estimates calculated from ten implicates of the data edited and imputed using the Bayesian method described in Section VI of the paper. All statistics in this Table are weighted using ASM sample weights for the observations in the given sample. Column 1 shows the within-R^2 from a regression of the (natural) log of the value of shipments on logged annual payroll and industry fixed effects. Column 2 is the within-R^2 from a regression of logged shipments on logged payroll, logged cost of materials, the log of capital, and industry fixed effects. Column 3 shows the standard deviation of the ratio of plant-level shipments over the plant-level capital stock, multiplied by the industry-level share of capital's cost as a share of total costs, scaled by the industry-level weighted average of this ratio. Columns 4 and 5 show the analogous standard deviations for the ratio of shipments over labor costs (payroll) and shipments over the cost of materials.

Table 4: The Effects of Data Cleaning on Modeled Within-Industry Productivity Dispersion and Allocative Efficiency Estimates, U.S. Annual Survey of Manufactures, 2009-2018

Data Version	N	$ln(\widetilde{TFPR})$ (1)	OP within (2)	AE (3)
Captured	188,000	0.92	1.17	0.01
Final	364,000	0.43	1.11	0.15
Final*	188,000	0.44	1.12	0.13
Bayesian (mean)	366,000	0.51	1.13	0.07
Bayesian (s.d.)	366,000	0.03	0.02	0.02

Note: The table shows cross-sectional modeled statistics from different versions of the 2009-2018 ASM data. The rows follow the same order as in Table 3. All statistics in this Table are weighted using ASM sample weights for the observations in the given sample. Column 1 shows the standard deviation of the natural log of plant-level revenue total factor productivity (TFPR), scaled by industry-level TFPR, which is itself the revenue-weighted harmonic mean of plant-level TFPR. Column 2 shows the "within" term from an (Olley and Pakes, 1996) decomposition of aggregate productivity. This is the sample-weighted average of plant-level TFPR as a share of revenue-weighted average TFPR. Column 3 shows cross-sectional Allocative Efficiency as defined by Bils et al. (2021). See text for details.

Table 5: The Effects of Data Cleaning on Panel R^2 and Dispersion of Plant-level Growth Rates, U.S. Annual Survey of Manufactures, 2009-2018

Data Version	N	Y_t (1)	M_t (2)	K_t (3)	L_t (4)	$TFPR_t$ (5)
Panel A: Regression within-R^2						
Captured	122,500	0.87	0.79	0.87	0.73	0.31
Final	273,000	0.93	0.90	0.97	0.92	0.56
Final*	122,500	0.94	0.90	0.97	0.94	0.56
Bayesian (mean)	273,000	0.90	0.85	0.81	0.90	0.31
Bayesian (s.d.)	273,000	0.02	0.04	0.03	0.02	0.09
Panel B: S.D. of Annual Growth Rates						
Captured	122,500	0.30	0.49	0.29	0.32	
Final	273,000	0.29	0.41	0.16	0.27	
Final*	122,500	0.26	0.39	0.17	0.23	
Bayesian (mean)	273,000	0.36	0.52	0.55	0.32	
Bayesian (s.d.)	273,000	0.06	0.08	0.06	0.05	

Note: Panel A shows within-R^2s from regressions of the natural log of each of five plant-level variables on their own one-year-lagged values and industry fixed effects: the value of shipments (Y), cost of materials (M), capital (K), annual payroll (L), and revenue total factor productivity (TFPR). Panel B shows the standard deviation of the plant-level annual growth rates of each of the first four variables (shipments, materials, capital and payroll). All statistics in this Table are weighted using ASM sample weights for the observations in the given sample.

Table 6: The Effects of Data Cleaning on Aggregate Productivity Growth Decompositions, U.S. Annual Survey of Manufactures, 2009-2018

Data Version	N	PL TE (1)	PL RE (2)	APG (3)
Captured	122,500	0.42	0.50	1.05
Final	273,000	-0.01	-0.02	-0.04
Final*	122,500	-0.01	-0.02	-0.03
Bayesian (mean)	273,000	-0.03	-0.01	-0.04
Bayesian (s.d.)	273,000	0.02	0.01	0.02

Note: The table shows the Petrin and Levinsohn (2012) aggregate productivity growth (APG) decomposition using different versions of the 2009-2018 ASM data. The rows follow the same order as in Table 3. All statistics in this Table are weighted using ASM sample weights for the observations in the given sample. Column 1 shows the contribution of plant-level technical efficiency growth to APG. Column 2 shows the contribution of the reallocation of inputs. The final column shows aggregate productivity growth as defined by Petrin and Levinsohn. All estimates are time-series averages of annual growth rates.

Appendix

A The Annual Survey of Manufactures and IRS Income and Payroll Tax Data

The **Annual Survey of Manufactures (ASM)** (U.S. Census Bureau, 2018) collects information on a rotating panel of establishments, with the largest establishments surveyed with certainty, and smaller plants in the frame with some probability (which is increasing with size). Samples are selected at 5-year intervals beginning in years ending with 4 or 9. Plants above a certain size —in 2009-2018, the threshold is 1000 employees —are sampled with certainty. Plants below that size threshold are selected with a probability that is a function of size, and remain in the sample until the end of the 5-year panel or the plant exits (whichever comes first). After the first year of the panel, the ASM is supplemented annually with a sample of entrants. In years ending in 2 or 7, the ASM is collected as a subsample of the Census of Manufactures (U.S. Census Bureau, 2017), which is part of the Economic Census. We use the ASM sample weights in all of our estimates in Tables 3-6.

To measure plant-level TFPR we use four plant-level variables: the total value of shipments, the total cost of materials (which includes costs of energy), total salaries and wages (i.e., annual payroll), and capital. In our imputation models (but not in our measure of plant-level productivity) we also use March 12 employment, because it is highly correlated with the other variables.

To measure capital in the final data, we use the real capital stock constructed as part of the BLS-Census Multifactor Productivity project, as described in Blackwood et al. (2023). We then multiply the real stock values by nominal rental rates (Kehrig, 2015) in order to measure the flow cost of capital services.[30] While Blackwood et al. (2023) provide separate measures of capital for equipment and structures, we use the plant-level sum of the two measures.

For the captured data, we construct our own capital stock measures using the perpetual inventory method applied to the raw data on reported values of assets and capital expenditures. To convert stocks to flow costs of capital services, we use the same interest rate that we use for the final data.

A.I Trimming the data

It is not uncommon for researchers to trim outliers, even in the already-cleaned final data. For instance, Hsieh and Klenow (2009) trim the tails of plant productivity and distortions in each country-year.[31] To aid comparison to the existing literature, for all results in Tables 3-6 we trim observations below the first percentile or above the 99th percentile of the (marginal) distribution of the marginal revenue product of capital, labor, or materials. (Before trimming, a given observation can be in the 1% tail of more than one marginal revenue product distribution.)

[30] Because the dispersion measures we are interested in are deviations from the industry average, whatever interest rate we pick cancels out and therefore does not affect our dispersion measures (as long as the chosen interest rate is constant within each industry).

[31] Bils et al. (2021) and Blackwood et al. (2021) only trim the productivity outliers, although (Bils et al. 2021 additionally trim plants with big changes and update the production function elasticities after trimming.

B A Theory of Misallocation

In this section, we briefly recap the theory of misallocation that has spawned a large literature on the topic (Bils et al., 2021; Hsieh and Klenow, 2009). In order to introduce the notation and the intuition, we focus on describing the baseline model used in (Bils et al., 2021). In a companion paper (Rotemberg and White, 2021b) we describe how the calculation changes when we include sector-specific demand elasticities, non-constant returns to scale in production, and roundabout production as in (Blackwood et al., 2021).

The wage, rental rate, and intermediate price are constant in the economy, but plants in each sector $s \in S$ face idiosyncratic distortions on each input.[32] As a result, each plant's profits are:[33] $\pi_{si} = P_{si}Q_{si} - \left(1+\tau_{L_{si}}\right) wL_{si} - \left(1+\tau_{K_{si}}\right) RK_{si} - (1+\tau_{M_{si}})p_M M_{si}$.

Demand is CES, so profit maximization implies that the plant's output price is a fixed markup over its marginal cost. Within a sector, the input share of each input will be proportional to the distortion, and there is complete pass-through of improvements in productivity A_{si} to prices. As a result, revenue productivity, $TFPR_{si} = \frac{P_{si}Q_{si}}{\left((rK)_{si}^{\alpha_s}(wL)_{si}^{1-\alpha_s}\right)^{\gamma_s}(p_M M)_{si}^{1-\gamma_s}}$, only varies due to the distortions. In particular,

$$TFPR_{si} = \frac{\sigma}{\sigma-1}\left[\left(\frac{\left(1+\tau_{K_{si}}\right)R}{\gamma_s\alpha_s}\right)^{\alpha}\left(\frac{\left(1+\tau_{L_{si}}\right)w}{\gamma_s(1-\alpha_s)}\right)^{1-\alpha}\right]^{\gamma_s}\left[\left(\frac{\left(1+\tau_{M_{si}}\right)p_M}{(1-\gamma_s)}\right)^{(1-\gamma_s)}\right]. \quad (1)$$

TFPQ can be inferred by taking advantage of the fact that the markup is known:

$$TFPQ_{si} \equiv A_{si} \propto \frac{\left(P_{si}Q_{si}\right)^{\frac{\sigma}{\sigma-1}}}{\left((RK)_{si}^{\alpha_s}(wL)_{si}^{1-\alpha_s}\right)^{\gamma_s}(p_M)^{1-\gamma_s}} \quad (2)$$

Sectoral productivity is a CES aggregator of TFPQ and (relative) TFPR,[34]

$$TFP_s = \left(\sum_{i\in s} A_{si}^{\sigma-1}\widetilde{TFPR}_{si}^{\,1-\sigma}\right)^{\frac{1}{\sigma-1}}. \quad (3)$$

Since we know from Equation 1 that $TFPR_{si}$ is equal to $\widetilde{TFPR}_s$ unless plants have idiosyn-

[32] We try to keep our notation standard - plant i in sector s produces quantity Q_{si} which it sells at a price of P_{si}. It uses three inputs, labor (L_{si}), capital (KL_{si}), and intermediate materials (ML_{si}), which respectively have prices w, r, and p_M. Plant si faces an exogenous multiplicative distortion for each input x $(1+\tau_{x_{si}})$ which generates a wedge between the national input price and the price paid by plant si.

[33] While we focus our attention on gross-output production functions, Hsieh and Klenow (2009) use a value added specification. Without roundabout production, there is no conceptual difference between materials and the other inputs.

[34] We define sectoral $\widetilde{TFPR}_s$ as the revenue-weighted harmonic mean of $TFPR_{si}$, and $\widetilde{TFPR}_{si} \equiv \frac{TFPR_{si}}{\widetilde{TFPR}_s}$.

cratic distortions, the "efficient" counterfactual sectoral TFP is $\overline{A}_s = \left(\sum_{i=1}^{M} A_{si}^{\sigma-1}\right)^{\frac{1}{1-\sigma}}$. The ratio of observed sectoral TFP to efficient sectoral TFP is our measure of allocative efficiency:

$$\widetilde{TFP_s} = \frac{TFP_s}{\overline{A}_s} = \left(\sum_{i \in s} \widetilde{TFPQ_{si}}^{\sigma-1} \widetilde{TFPR_{si}}^{1-\sigma}\right)^{\frac{1}{\sigma-1}}, \tag{4}$$

where $\widetilde{TFPQ}_{si} \equiv \frac{A_{si}}{\overline{A}_s}$. In this model, sectoral output is aggregated using a Cobb-Douglas production function, and there are no distortions between sectors. So aggregate allocative efficiency in this economy is given by

$$\widetilde{TFP} = \prod_{s \in S} \widetilde{TFP_s}^{\theta_s}, \tag{5}$$

where (θ_s) is the gross output share of sector s.

Instead of measuring how sensitive the measures of aggregate productivity are to different underlying economic assumptions, which has been the primary focus of much of the recent methodological literature on misallocation (Asker et al., 2014; Haltiwanger et al., 2018), we focus on how this measure of allocative efficiency is affected by different choices in data cleaning.

There are a few important parameters that are not directly reported in the data: the production function elasticities and the demand elasticity of substitution. We use cost shares for the former (and do not update the elasticities when using different cuts of the data), and 4 for the latter (Redding and Weinstein, 2020). Rotemberg and White (2021b) shows the results are similar using industry-specific elasticities of substitution (Ahmad and Riker, 2019).

C Petrin and Levinsohn Aggregate Productivity Growth Decomposition

Petrin and Levinsohn (2012) [PL] develop a continuous-time decomposition of aggregate productivity growth. Obviously the data collected by the ASM is for (discrete) annual intervals. We use Tornqvist-Divisia approximations to approximate the PL APG decomposition. The Tornqvist approximation for APG is:

$$APG_t = \sum_i \overline{D}_{it}^{v} \Delta lnVA_{it} - \sum_i \sum_k \overline{c}_{ikt} \Delta lnX_{ikt} \tag{6}$$

where:

- bars over variables denote two-year averages
- Δ is the first-difference operator
- D_{it}^{v} is plant i's value-added share in year t
- VA_{it} is plant i's value-added in year t

- c_{ikt} is plant i's expenditures for primary input k in year t divided by aggregate value added
- X_{ikt} is primary input k at plant i in year t

The discrete-time approximation of the PL APG decomposition is:

$$\begin{aligned} APG_{G,t} &= \sum_i \overline{D}_{it} \sum_k (\varepsilon_{I(i)k} - \overline{s}_{ikt}) \Delta ln X_{ikt} + \sum_i \overline{D}_{it} \sum_j (\varepsilon_{I(i)j} - \overline{s}_{ijt}) \Delta ln M_{ijt} \\ &\quad - \sum_i \overline{D}_{it} \Delta ln F_{it} + \sum_i \overline{D}_{it} \Delta ln \omega_{it}, \end{aligned} \tag{7}$$

where:

- D_{it} is the Domar weight of plant i in year t ($\frac{Y_{it}}{\sum_i VA_{it}}$)
- Y_{it} is the value of shipments from plant i in year t
- ε_{ik} is the output elasticity of primary input k in plant i's industry I
- s_{ikt} denotes primary input k's costs as a share of the value of shipments at plant i in year t
- X_{ikt} is primary input k at plant i in year t
- $\varepsilon_{I(i)j}$ is the output elasticity of intermediate input j at plant i in industry I
- s_{ijt} is the cost of intermediate input j as a share of the value of shipments at plant i in year t
- M_{ijt} is intermediate input j at plant i in year t
- F_{it} are fixed costs at plant i in year t
- ω_{it} is plant i's TFP, which PL calls technical efficiency, in year t

The first two summation terms on the right hand side of 7 are the contribution of the reallocation of primary and intermediate inputs, respectively, to aggregate productivity growth. The final summation term is the contribution of within-plant total factor productivity growth to APG. We do not observe changes in fixed costs F_{it} in our data directly, but we can estimate the aggregate fixed cost term as APG minus the aggregate reallocation and technical efficiency terms.

D Effects of Each Type of Edit on Measured Allocative Efficiency and Aggregate Productivity Growth

Table A.1 showed some of the most common types of Census Bureau edits and imputes in the ASM. These are not necessarily the most impactful edits/impute on measures of productivity dispersion or productivity growth. In Table A.2 we list the share of observations that are affected by each of nine categories of edits or imputes that have significant effects on productivity dispersion and/or aggregate productivity growth decompositions.

Determining the effect of a single category of edits/imputes on aggregate productivity or aggregate productivity growth is not a simple task. A given observation can be edited or imputed in multiple ways, because edits are applied separately to different variables. Analogous to the way workers in a plant interact with each other to affect productivity, edits and imputes in a dataset can interact with each other to affect statistics of interest. Table A.3 shows two alternative sets of estimates of how much each of the categories in Table A.2 affects each of three statistics from the main Tables: (1) the Bils et al. (2021) measure of allocative efficiency (AE), (2) aggregate productivity growth (APG), and (3) the contribution of reallocation to APG in the Petrin-Levinsohn decomposition. For each of these statistics, we calculate two decompositions. The first decomposition (labeled "first" in the Table) is the outcome of the following experiment. We calculate the statistic of interest (e.g., allocative efficiency) twice: first using just the captured data, and then using the final data for the observations which had the "flow variable missing" flag and the captured data for the rest of the observations. The difference between these two estimates of allocative efficiency is an estimate of how much the "flow variable missing" category of imputes changed the estimated value of allocative efficiency. Then we repeat the same procedure for each of the other eight categories of edits/imputes listed in the first column of the Table. Finally, we divide each edit/impute category's contribution by the total change in the value of the allocative efficiency statistic when going from the dataset with only captured data to the dataset with all final data. These shares are presented in the "first" columns of Table A.3 (columns 1, 3, and 5).

The order in which changes are made to the data (going from captured to final) matters for how much those changes affect the statistic of interest. To avoid this path-dependence problem, we calculate the calculate Shapley (1953) value of each category of edit/imputation; this is the average contribution of an edit/impute category to the overall change in the statistics, averaged over every permutation of the order of the edit/impute categories. The columns in Table A.3 labeled "Shapley" (columns 2, 4, and 6) show each edit/impute category's Shapley value as a share of the total change in the statistic (from captured to final data). The effects of the edit/impute categories differ across statistics. For example, regression edits for materials have a large effect on the change in measured allocative efficiency—38% of the overall change—but a small effect on measured aggregate productivity growth (-4%).[35] However, across all three statistics in both the first and the Shapley decompositions, analyst edits account for a large share of the overall change in the statistic. In the case of aggregate productivity growth, analyst edits account for 90% of the overall change going from captured to final data.

D.I Detailed Definitions of Edit/Impute Categories

In Table A.2 we provide brief definitions of the edit/impute categories used in Table A.3. Here we provide more detailed definitions of each of the categories, in the same order as in the Tables. The category "flow variable missing" applies to the value of shipments,

[35] A given edit can make a negative contribution to the overall change if, for example, the overall change is negative, and the given edit has a positive effect on the change in the statistic.

the cost of materials, and or annual payroll. We use it whenever the final datum is not missing but the captured datum is missing.[36]

We split out logical edits for payroll, shipments, and materials. Logical edits are made when there are multiple survey questions asking for the same information. For example, the ASM asks respondents for the plant's production worker wages, non-production worker wages, and total salaries and wages (which should be the sum of the first two). To give a simple example, if reported non-production worker wages are, e.g., only half of the reported total salaries and wages, and both are plausible for the given plant in the given industry, but non-production worker wages were not reported, then the Bureau will impute non-production worker wages for this plant as total salaries and wages minus production worker wages.

Regression edits (which we report for materials) are used to edit data when alternative sources of information are not available. The U.S. Census Bureau uses a variety of industry-specific regression-based imputation strategies. Since they do not require any observed alternative value, regression imputes are often used to impute missing values when no other information is available for a given variable, preventing the use of logical-type imputes. In general, regression edits create predictions using one other variable, and (for plants surveyed in two consecutive years of the ASM), one-year lags of the that variable and the dependent variable. For example, for a plant in two consecutive years of the ASM, cost of materials would be imputed as the predicted value from a regression of materials on total value of shipments (TVS) and one-year lags of TVS and the cost of materials. The regression coefficients used in this imputation are estimated on a sample of plants for which all four variables are reported in the same year and industry as the plant with cost of materials that is deemed to be in error. Unlike administrative and logical edits, there is not necessarily any direct evidence that the value reported by the establishment may be incorrect. The Census Bureau uses the ratio bounds we use in Section VI to flag reported values to be replaced with imputes.

Analyst edits rely on the expertise of full-time industry specialists employed by the U.S. Census Bureau. Note that analysts can try to use alternative sources (such as an internet search) to verify or edit values.

The divide-by-1000 edits happen when the ratio of a dollar-valued variable (e.g., annual payroll) over employment is 1000 times greater than what is typical for the plant's industry. In these cases the Census Bureau suspects that the respondent answered in dollar units even though the questionnaire asks for values in thousands of dollars.

Administrative records ("ad rec") edits are similar to logical edits, but differ in that the alternative source of information is from administrative records. The administrative records come from IRS payroll tax and IRS income tax data. In our ASM sample these are currently only used for payroll (total wages) or shipments. For annual payroll, the concept measured in the ASM is (by design) the same as the payroll concept measured on the IRS Form 941. Although a measure of revenue from income tax data is sometimes used to

[36] There are some observations with missing capital data but where the corresponding flag does not correspond to impute missing; we overrule those flags for this categorization.

impute for shipments in the ASM, conceptually they are not the same thing. In particular, shipments from one plant to another belonging to the same firm count as shipments in the ASM, but they would not necessarily contribute to revenue on IRS income tax forms. There are some administrative data for assets and employment, but the Census Bureau does not use them for imputation in our ASM sample. The Bureau does not have access to administrative data to impute for the cost of materials.

We manually categorize the residual changes into ones that are replicable and ones that are not. These include regression-types of edits that we do not categorize in the regression edits for materials (such as regressions for the other variables, or directly imputing the midpoint of the ratio bound).

E Bayesian Simultaneous Edit-Imputation

In this section, we describe the details of the Kim et al. (2015) algorithm and how we implement the approach for our real data analysis. First, we define notations including those introduced in the main text. Establishment i reports p characteristics, $\boldsymbol{y}_i = \{y_{i1}, y_{i2} \dots y_{ip}$ (where items could be missing). The corresponding true values are $\boldsymbol{x}_i = \{x_{i1}, x_{i2} \dots x_{ip}$, with an underlying distribution f_X. If establishment i did not report characteristic j or reported an incorrect value, indicator s_{ij} has the value one, i.e., $s_{ij} = 0$ if $y_{ij} = x_{ij}$. We define the the feasible region $\mathcal{D}$ of plausible reports and set $s_{ij} = 0$ for all j's if $\boldsymbol{y}_i \in \mathcal{D}$, i.e., the reported values of establishment i satisfy all the edit constraints. [37]

E.I Nonparametric Bayesian edit-imputation method

Given the dataset of reported values $\boldsymbol{Y} = \{\boldsymbol{y}_1, \boldsymbol{y}_2, \dots \boldsymbol{y}_n\}$, the goal of data cleaning is finding which characteristics are likely to be incorrect, then replacing those with estimated values of the true values $\hat{x}_{ij}$. The Kim et al. (2015) method assumes a Bayesian hierarchical model with three levels for the triplet $(\boldsymbol{x}_i, \boldsymbol{y}_i, \boldsymbol{s}_i)$.

First, the distribution of true values is assumed with the Dirichlet process (DP) Gaussian mixture model (Ishwaran and James, 2001). In the manufacturing data, the underlying distribution f_X is heavily skewed and has complex features, which are not easily captured by standard multivariate distributions. We adopt the Gaussian mixture model (MacEachern and Müller, 1998) modified for the imputation and editing Kim et al. (2014, 2015) with the feasible region $\mathcal{D}$: for $i = 1, \dots, n$ and $k = 1, \dots, K$

$$
\begin{aligned}
\boldsymbol{x}_i | \mu, \Sigma, z_i, \mathcal{D} &\sim N(\log \boldsymbol{x}_i | \boldsymbol{\mu}_{z_i}, \boldsymbol{\Sigma}_{z_i})\ I\left[\boldsymbol{x}_i \in \mathcal{D}\right], \\
z_i &\sim \text{Categorical}(\pi_1, \dots, \pi_K), \\
\boldsymbol{\mu}_k | \boldsymbol{\Sigma}_k &\sim N\left(\bar{\boldsymbol{\mu}}_0, \frac{1}{h_0}\boldsymbol{\Sigma}_k\right) \\
\boldsymbol{\Sigma}_k &\sim \text{InverseWishart}(\zeta_0, \boldsymbol{\Phi}),
\end{aligned} \tag{8}
$$

where $\mu = (\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K)$, $\Sigma = (\boldsymbol{\Sigma}_1, \dots, \boldsymbol{\Sigma}_K)$, $\boldsymbol{\Phi} = \text{diag}(\Phi_1, \cdots, \Phi_p)$, and the membership indicator z_i denotes which mixture component (out of K) $\boldsymbol{x}_i$ belongs to. As the DP prior of

[37] However, analyst's editing can follow this automatic edit-imputation, so $\boldsymbol{y}_i \in \mathcal{D}$ is also further edited.

the membership probabilities π_k, we adopt the stick-breaking representation of Sethuraman (1994), so the first few mixture components receive relatively high prior probabilities π_k than the latter components:

$$\begin{aligned} \pi_k &= v_k \prod_{g<k}(1-v_g) \\ v_k &\sim \text{Beta}(1,\alpha) \ \text{ for } \ k=1,\ldots,K-1; \ \ v_K = 1, \\ \alpha &\sim \text{Gamma}(a_\alpha, b_\alpha) \end{aligned} \tag{9}$$

where $\text{Gamma}(a,b)$ denotes the gamma distribution of shape parameter a and rate parameter b whose mean is a/b. As in Kim et al. (2014) and Kim et al. (2015), we added a pre-determined, small number to $x_i = 0$ in the original data before putting the logarithm to facilitate computation and prior specification.

Recall that the flag indicator is defined as $s_{ij} = 0$ if $y_{ij} = x_{ij}$ and 1 if y_{ij} is missing or determined to have an erroneous value. Given $\boldsymbol{s}_i$, we partition the reported values as $\boldsymbol{y}_i = (\boldsymbol{y}_{i,U}, \boldsymbol{y}_{i,F})$ where $\boldsymbol{y}_{i,U} = \{y_{ij} : s_{ij} = 0, j = 1, \cdots, p\}$ denotes the "unflagged" items and $\boldsymbol{y}_{i,F} = \{y_{ij} : s_{ij} = 1, j = 1, \cdots, p\}$ and $\boldsymbol{x}_i = (\boldsymbol{x}_{i,U}, \boldsymbol{x}_{i,F})$ in the same manner. Then, the distribution of reported values $\boldsymbol{y}_i$ is expressed with the flag indicator:

$$\begin{aligned} f(\boldsymbol{y}_i|\boldsymbol{x}_i, \boldsymbol{s}_i, \mathcal{D}) &= f(\boldsymbol{y}_{i,F}|\mathcal{D}, \boldsymbol{s}_i, \boldsymbol{y}_{i,U})\delta(\boldsymbol{y}_{i,U} - \boldsymbol{x}_{i,U}), \\ f(\boldsymbol{y}_{i,F}|\mathcal{D}, \boldsymbol{s}_i, \boldsymbol{y}_{i,U}) &\propto \prod_{\{j:s_{ij}=1\}} I[L_j \le y_{ij} \le U_j] \cdot I[\boldsymbol{y}_i \notin \mathcal{D}], \end{aligned} \tag{10}$$

where $I[\cdot] = 1$ if the statement is true and 0 otherwise, and $\delta(\cdot)$ is the Dirac delta function with the point mass at zero. The measurement error model of $\boldsymbol{y}_{i,F}$ assumes that the reporeted values are not in the feasible region $\mathcal{D}$, i.e., at least an edit constraint is violated. However, for computational feasibility, the analysts need to set bounds (L_j, U_j) for any reported values, for example, using the historical data. Note that we implicilty assume the uniform prior distribution for the location of incorrect measurements $\boldsymbol{y}_{i,F}$. One may put other distributional assumption for the measurement errors, for example, the normal error $y_{ij} - x_{ij} \sim N(0, s_j^2)$. However, in our application example, the patterns of measurement errors are too complicated to be formulated with a specific distributional assumption. Therefore, we let the data find the measurement error pattern rather than putting a closed form of the prior distribution.

Lastly, we set $\phi_j \sim \text{Gamma}(0.25, 0.25)$ and the following hyperparameter values as used in Kim et al. (2015): $\bar{\boldsymbol{\mu}}_0 = \boldsymbol{0}$, $\zeta_0 = p + 1$, and $a_\alpha = b_\alpha = 0.25$. We refer readers to Kim et al. (2015) for the rationale and more details for the hyperparameter setting.

E.II Markov chain Monte Carlo steps

With the Bayesian model introduced in the previous subsection, estimation of all model parameters and true values of missing or erroneous reported values is made by their posterior distribution via the Markov chain Monte Carlo (MCMC). This section presents the details of the full MCMC implementation.

The following steps are repeated for enough number of iterations. Let the current iteration number be t. We put q in the superscript for a proposed value (e.g., x_{ij}^q) and omit the superscript $(t-1)$ or (t) for the current values updated at the previous or current iteration (e.g., x_{ij} instead of $x_{ij}^{(t-1)}$).

1. For each edit-failing record i such that $\boldsymbol{y}_i \notin \mathcal{D}$, update $\boldsymbol{s}_i = (s_{i1}, \ldots, s_{ip})$ and $\boldsymbol{x}_i = (x_{i1}, \ldots, x_{ip})$.

   (a) Find all n_{death} possible "death" procedures $\boldsymbol{s}_i^* = (s_{i1}^*, \ldots, s_{ip}^*)$ such that $\sum_{j=1}^p s_{ij}^* = \sum_{j=1}^p s_{ij} - 1$. It means that a reported value declared as faulty at iteration $t-1$ is now considered as correct.
   Similarly, find all n_{birth} possible "birth" procedures $\boldsymbol{s}_i^*$ such that $\sum_{j=1}^p s_{ij}^* = \sum_{j=1}^p s_{ij} + 1$. It means that a reported value declared as correct at iteration $(t-1)$ is now considered as faulty.
   Then, with the probability $1/2$, randomly choose whether a death or birth procedure is selected. Then, randomly choose a death (or birth) process $\boldsymbol{s}_i^q$ out of n_{death} possible death procedures (or n_{birth} possible birth procedures) with the probability $g(\boldsymbol{s}_i^q|\boldsymbol{s}_i) = 1/n_{\text{death}}$ (or $1/n_{\text{birth}}$).

   (b) Given the proposed flag indicator $\boldsymbol{s}_i^q$, propose $\boldsymbol{x}_i^q = (\boldsymbol{x}_{i,F}^q, \boldsymbol{x}_{i,U}^q)$ with the following procedure. By the definition of s_{ij}^q, $\boldsymbol{x}_{i,U}^q = \boldsymbol{y}_{i,U}^q = \{y_{ij}^q : s_{ij}^q = 0, j = 1, \cdots, p\}$. For $\boldsymbol{x}_{i,F}^q = \{x_{ij}^q : s_{ij}^q = 1, j = 1, \cdots, p\}$ which are the estimated true values corresonding to the flagged reported values, propose their log-transformed values from the conditional normal distribution $g(\boldsymbol{x}_{i,F}^q|\mu, \Sigma, z_i, \boldsymbol{s}_i^q)$ that is $\log \boldsymbol{x}_{i,F}^q \sim N(\tilde{\boldsymbol{\mu}}_{i,F}^q, \tilde{\boldsymbol{\Sigma}}_{i,F}^q)$ such that $\tilde{\boldsymbol{\mu}}_{i,F}^q = \boldsymbol{\mu}_{i,F} + \boldsymbol{\Sigma}_{i,FU}\boldsymbol{\Sigma}_{i,U}^{-1}(\boldsymbol{x}_{i,U} - \boldsymbol{\mu}_{i,U})$ and $\tilde{\boldsymbol{\Sigma}}_{i,F}^q = \boldsymbol{\Sigma}_{i,F} - \boldsymbol{\Sigma}_{i,FU}\boldsymbol{\Sigma}_{i,U}^{-1}\boldsymbol{\Sigma}_{i,UF}$ where the mean vector and the covariance matrix at iteration $(t-1)$ are sorted with the values of s_{ij}^q and partitioned as

$$\boldsymbol{\mu}_{z_i} = \begin{pmatrix} \boldsymbol{\mu}_{i,F} \\ \boldsymbol{\mu}_{i,U} \end{pmatrix} \left(\text{and } \boldsymbol{\Sigma}_{z_i} = \begin{pmatrix} \boldsymbol{\Sigma}_{i,F} & \boldsymbol{\Sigma}_{i,FU} \\ \boldsymbol{\Sigma}_{i,UF} & \boldsymbol{\Sigma}_{i,U} \end{pmatrix} \right($$

   (c) Accept the proposed values $(\boldsymbol{x}_i^q, \boldsymbol{s}_i^q)$ with the acceptance proability

$$\min \left\{ 1, \left(\frac{f(\boldsymbol{y}_{i,F}|\mathcal{D}, \boldsymbol{s}_i^q, \boldsymbol{y}_{i,U}) f(\boldsymbol{x}_i^q|\mu, \Sigma, z_i, \mathcal{D})}{f(\boldsymbol{y}_{i,F}|\mathcal{D}, \boldsymbol{s}_i, \boldsymbol{y}_{i,U}) f(\boldsymbol{x}_i|\mu, \Sigma, z_i, \mathcal{D})} \frac{g(\boldsymbol{x}_{i,F}|\mu, \Sigma, z_i, \boldsymbol{s}_i) g(\boldsymbol{s}_i|\boldsymbol{s}_i^q)}{g(\boldsymbol{x}_{i,F}^q|\mu, \Sigma, z_i, \boldsymbol{s}_i^q) g(\boldsymbol{s}_i^q|\boldsymbol{s}_i)} \right) \right\} \left($$

2. For each i such that $\boldsymbol{y}_i \notin \mathcal{D}$, update $\boldsymbol{x}_i = (x_{i1}, \ldots, x_{ip})$ only where $\boldsymbol{s}_i$ is held fixed.

   (a) The proposal step is the same with that in Step 1.(b) except $\boldsymbol{s}_i$ is fixed, i.e., propose $x_{i,F}^q$ from $f(\boldsymbol{x}_{i,F}^q|\mu, \Sigma, z_i, \boldsymbol{s}_i)$ that is $\log \boldsymbol{x}_{i,F}^q \sim N(\tilde{\boldsymbol{\mu}}_{i,F}^q, \tilde{\boldsymbol{\Sigma}}_{i,F}^q)$ such that $\tilde{\boldsymbol{\mu}}_{i,F}^q =$

$\boldsymbol{\mu}_{i,F} + \boldsymbol{\Sigma}_{i,FU}\boldsymbol{\Sigma}_{i,U}^{-1}(\boldsymbol{x}_{i,U} - \boldsymbol{\mu}_{i,U})$ and $\tilde{\boldsymbol{\Sigma}}_{i,F}^{q} = \boldsymbol{\Sigma}_{i,F} - \boldsymbol{\Sigma}_{i,FU}\boldsymbol{\Sigma}_{i,U}^{-1}\boldsymbol{\Sigma}_{i,UF}$ where the mean vector and the covariance matrix at iteration $(t-1)$ are sorted with the values of s_{ij} and partitioned as

$$\boldsymbol{\mu}_{z_i} = \begin{pmatrix} \boldsymbol{\mu}_{i,F} \\ \boldsymbol{\mu}_{i,U} \end{pmatrix} \left(\text{and } \boldsymbol{\Sigma}_{z_i} = \begin{pmatrix} \boldsymbol{\Sigma}_{i,F} & \boldsymbol{\Sigma}_{i,FU} \\ \boldsymbol{\Sigma}_{i,UF} & \boldsymbol{\Sigma}_{i,U} \end{pmatrix} \right($$

(b) Accept the proposed value $\boldsymbol{x}_i^q$ with the acceptance proability

$$\min \left\{ 1, \frac{f(\boldsymbol{x}_i^q|\mu,\Sigma,z_i,\mathcal{D})}{f(\boldsymbol{x}_i|\mu,\Sigma,z_i,\mathcal{D})} \frac{g(\boldsymbol{x}_{i,F}|\mu,\Sigma,z_i,\boldsymbol{s}_i)}{g(\boldsymbol{x}_{i,F}^q|\mu,\Sigma,z_i,\boldsymbol{s}_i)} \right\}.$$

3. For each $i = 1,\ldots,n$, update z_i from $z_i \sim \text{Category}(\pi_{i1}^*,\ldots,\pi_{iK}^*)$ where

$$\pi_{ik}^* = \frac{\pi_k N(\log \boldsymbol{x}_i|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)}{\sum_{k'=1}^K \pi_{k'} N(\log \boldsymbol{x}_i|\boldsymbol{\mu}_{k'},\boldsymbol{\Sigma}_{k'})} \text{ for } k = 1,\ldots,K.$$

4. For each $k = 1,\ldots,K$, update $(\boldsymbol{\Sigma}_k,\boldsymbol{\mu}_k)$ with its conjugate forms. Let $n_k = \sum_{i=1}^n I[z_i = k]$, $\bar{\boldsymbol{x}}_k = \left(\sum_{i=1}^n I[z_i = k]\log \boldsymbol{x}_i\right)/n_k$, and $\boldsymbol{S}_k = \sum_{i=1}^n I[z_i = k](\log \boldsymbol{x}_i - \bar{\boldsymbol{x}}_k)(\log \boldsymbol{x}_i - \bar{\boldsymbol{x}}_k)^\top$. Update

$$\boldsymbol{\Sigma}_k \sim \text{InverseWishart}\left(\zeta_0 + n_k, \boldsymbol{\Phi} + \boldsymbol{S}_k + \frac{h_0 n_k}{h_0 + n_k}(\bar{\boldsymbol{x}}_k - \bar{\boldsymbol{\mu}}_0)(\bar{\boldsymbol{x}}_k - \bar{\boldsymbol{\mu}}_0)^\top\right)\left($$

then

$$\boldsymbol{\mu}_k \sim N\left(\frac{h_0\bar{\boldsymbol{\mu}}_0 + n_k\bar{\boldsymbol{x}}_k}{h_0 + n_k}, \frac{1}{h_0 + n_k}\boldsymbol{\Sigma}_k\right).$$

5. For each $k = 1,\ldots,K-1$, update $v_k \sim \text{Beta}(1 + n_k, \alpha + \sum_{g>k} n_g)$, and let $v_K = 1$. Then, calcuate the mixture component weights $\pi_k = v_k \prod_{g<k}(1 - v_g)$.

6. For each $j = 1,\ldots,p$, update $\Phi_j \sim \text{Gamma}(a_\Phi + K\zeta_0/2, b_\Phi + \Sigma_{k(j,j)}^{-1}/2)$ where $\Sigma_{k(j,j)}^{-1}$ is the j-th diagonal element of $\boldsymbol{\Sigma}_k^{-1}$.

7. Update $\alpha \sim \text{Gamma}(a_\alpha + K - 1, b_\alpha - \log \pi_K)$.

E.III Implemenataion of the Bayesian edit-imputation algorithm

In our implementation we define the feasible region of plausible reports $\mathcal{D}$ using only a set of ratio edit rules which bound the ratios of any two variables.[38] The ratio edit rules can come either from industry specific knowledge, or from outliers in the data itself.

[38] In addition to ratio edit rules, the Kim et al. (2015) algorithm is designed to also use balance edits which require entries to add up. For instance, total wages = production worker wages + non-production worker

Fellegi and Holt (1976) note that the set of explicit ratio edit rules can imply additional ones as well.[39] While $\boldsymbol{s}_i$ is not directly observed, A_i indexes the failed ratio edit rules.[40] If, e.g., y_{i1} fails multiple edits and y_{i2} fails only one, then, other things equal, y_{i1} is more likely to be faulty than y_{i2}.

For the baseline Bayesian model, we use the relevant ratios that are explicitly bounded (above and below) by the Census Bureau: $\frac{shipments}{materials}$, $\frac{employment}{wages}$ and $\frac{shipments}{wages}$.

The Bayesian model is very much driven by the data, and requires setting only a few parameters in the R code (which is available on the FSRDC server and will be made available in our replication package when this paper is accepted for publication in a peer-reviewed journal). Following Kim et al. (2015), we set the maximum possible number of components distributions, K. We set K=20, which is large enough that no data are in the lowest probability components in any industry-year. For the Markov Chain Monte Carlo, we choose a burn-in of 2000 iterations, which is long enough that even the largest industries' distributions appear to converge and have good mixing properties.[41] We then run the chain another 5,000 iterations, keeping every 500th iteration so that we have 10 completed datasets ("implicates"). Other than telling the algorithm the folder structure, which variables to use, and what ratio and balance edits the variables are subject to, no other choices need to be made by the modeller.

There is one caveat which applies to any edit-imputation method that uses ratio edits, including the Census Bureau's methods. In the ASM, the questionnaires ask the respondent to report in thousands of dollars. In some cases, the respondents appear to report in dollars. The Census Bureau determines this by looking at the ratio of each the dollar-valued variables over employment. If for every dollar-valued variable X, a plant's ratio of $\frac{X}{employment}$ fails the ratio edit, but $\frac{X/1000}{employment}$ satisfies the edit, then the Bureau replaces the dollar-valued variable X with $\frac{X}{1000}$. Note that if the Census Bureau did not do this, its Fellegi and Holt (1976) algorithm would choose to replace the employment value for this plant and keep all the (very large) reported dollar valued variables, since that would minimize the number of changes to the data. Before feeding the U.S. captured data into the Bayesian edit-imputation algorithm, we replace the captured value with the final value for any variable that was edited by the Census Bureau using this divide-by-1000 edit.

F Appendix Tables

wages, or more generally $\left(x_{iT_\ell} - \sum_{j \in \beta_\ell} x_{ij} = 0\right)$ for x_{iT_ℓ} as the total for the ℓth balance rule for the set of component variables β_ℓ. In their application of the algorithm to one industry in the 2007 CMF, Kim et al. (2015) use 12 variables that are subject to ratio edits, six of which are totals subject to balance edits, and 15 component variables that are subject to balance edits but not ratio edits.

[39] For instance, rules $x_1 \leq x_2$ and $x_2 \leq x_3$ imply $x_1 \leq x_3$.

[40] If we were using balance edits in our implementation, A_i would also indicate failed balance edit rules.

[41] Note that each iteration for a given industry-year involves taking draws for all of the missing and faulty data until all the observations we are modelling have been filled in.

Table A.1: Types of Edits and Imputations in U.S. Annual Survey of Manufactures

Edit/Impute Flag	Occurs When...
Administrative (A)	The item is imputed by direct substitution of corresponding administrative data (for the same establishment/record).
Cold Deck Statistical (B)	The item is imputed from a statistical (regression/beta) model based on historic data.
Analyst Corrected (C)	The reported value fails an edit, and an analyst directly corrects the value.
Model (Donor) Record (D)	The item is imputed using hot deck methods.
Receipts (F)	Receipts posted from the nonemployer universe as a result of the Business Master File EIN/SSN cross-reference match or primary SSN (one-for-one match).
High/Low (E)	The item is imputed with a value near the endpoints of the imputation range.
Historic (H)	The item is imputed by using historic ratio data for the same establishment.
Subject Matter Rule (J)	The item is imputed using a subject matter defined rule (e.g. y=1/2x).
Prior Year Ratio (HQ)	The item is imputed using a ratio of historic data times current reported values.
Raked (K)	The sum of a set of detail items do not balance to the total. The details are then changed proportionally to correct the imbalance.
Logical (L)	The item's imputation value is defined by an additive mathematical relationship (e.g., obtaining a missing detail item by subtraction).
Midpoint (M)	The item is imputed by direct substitution of midpoint of imputation range.
Rounded (N)	The reported value is replaced by its original value divided by 1000.
Prior Year Administrative (P)	The item is imputed by ratio imputation using corresponding administrative data from prior year (for same establishment).
Direct Substitution (S)	The item is imputed by direct substitution of another item's value (from within the same questionnaire.)
Trim-and-Adjusted (T)	The item was imputed using the Trim-and Adjust balancing algorithm.

Continued on next page

Table A.1 continued from previous page

Edit/Impute Flag	Occurs When...
Unable to Impute (U)	The reported item is blank or fails an edit, and the system cannot successfully substitute a statistically reasonable value for the original data.
Industry Average (V)	The item is imputed by ratio imputation using an industry average.
Warm Deck Statistical (W)	The item is imputed from a statistical (regression/beta) model based on current data.
Unusable (X)	The sum of a set of detail items cannot be balanced to the total because none of the scripted solutions achieved a balance.
Data Impute (2)	Data item imputed from a reported data item
Payroll Quarterly (4)	First quarter payroll reported higher than annual payroll; data adjusted during general data prep or legacy edits

Note: The table shows the types of edit/imputation flags used in the the U.S. Annual Survey of Manufactures from Grim (2011) and White et al. (2018), with a few additional categories. These flags are available to all researchers with access to the confidential U.S. manufacturing data used in this paper. The flags explain how the data was changed when the final data disagree with the captured data.

Table A.2: Important Categories of Census Bureau Edits/Imputes in the ASM

Edit/Impute Category	Share	Underlying Flags
Flow variable missing	0.303	Any of the variables shipments, payroll, or materials is missing.
Logical edit for materials	0.009	Materials flag = RL
Logical edit for shipments	0.017	Shipments flag = RL
Logical edit for payroll	0.029	Payroll flag = RL
Regression edit for materials	0.020	Materials flag = RB or RW
Analyst edit	0.090	Any variable's flag = RC
Divide-by-1000 edit	0.005	Any variable's flag = RN
Any ad rec edit	0.019	Either reported payroll or reported shipments was replaced with administrive records data or an impute using administrative records data
Other replicable edit	0.008	Reported data for shipments, payroll, or materials is replaced by an edit or impute that is theoretically replicable in other data sets and is not covered by one of the flags specified above.

Note: The table shows 9 categories of Census Bureau edits and imputations in the 2009-2018 Annual Survey of Manufactures and the share of observations that have at least one of these flags set. With a few exceptions—for example, "logical edit for materials" and "regression edit for materials" are mutually exclusive—a plant can be in more than one of these categories.

Table A.3: Decomposition of Effects of Census Bureau Edits/Imputations on Measured Allocative Efficiency and Aggregate Productivity Growth 2009-2018 Annual Survey of Manufactures

Edit/Impute Category	AE		APG		Reallocation	
	first	Shapley	first	Shapley	first	Shapley
	(1)	(2)	(3)	(4)	(5)	(6)
Flow variable missing	0.10	0.06	0.15	0.11	0.14	0.11
Logical edit for shipments	0.01	0.01	0.02	-0.07	0.02	-0.03
Logical edit for payroll	0.04	0.02	0.09	0.04	0.09	0.07
Logical edit for materials	-0.01	0.07	0.08	-0.03	0.10	0.04
Regression edit for materials	0.18	0.38	0.04	-0.04	0.05	0.01
Analyst edit	0.60	0.41	0.35	0.90	0.29	0.58
Divide-by-1000 edit	0.05	0.03	0.12	0.02	0.14	0.09
Any ad rec edit	-0.00	0.01	0.11	0.10	0.11	0.11
Other replicable edit	0.02	0.01	0.05	-0.04	0.06	0.02

Note: The table shows two alternative sets of estimates of how much each of the categories in Table A.2 affects each of three statistics from the main tables. Columns (1) and (2) show decompositions of the effects on the Bils et al. (2021) measure of allocative efficiency. Columns (3) and (4) show the effects on aggregate productivity growth. Columns (5) and (6) show the effects on the contribution of reallocation to APG in the Petrin-Levinsohn decomposition. Within each pair of columns, the column labeled 'first' shows the share of the overall change in the statistic attributed to each edit/impute category when we calculate the effect of those edits/imputes first without any other edits. Within in each pair of columns, the column labeled 'Shapley' shows the Shapley value of that type of edit/imputation as a share of the total change in the given statistic in going from all captured data to all final data.